

Bayesian Networks

Probabilistic Reasoning, Real-World Applications & Use in LLMs

Probability Theory

Graphical Models

Machine Learning

AI Inference

What Is a Bayesian Network?

A Bayesian Network is a probabilistic graphical model that represents variables and their conditional dependencies via a Directed Acyclic Graph (DAG).

Nodes

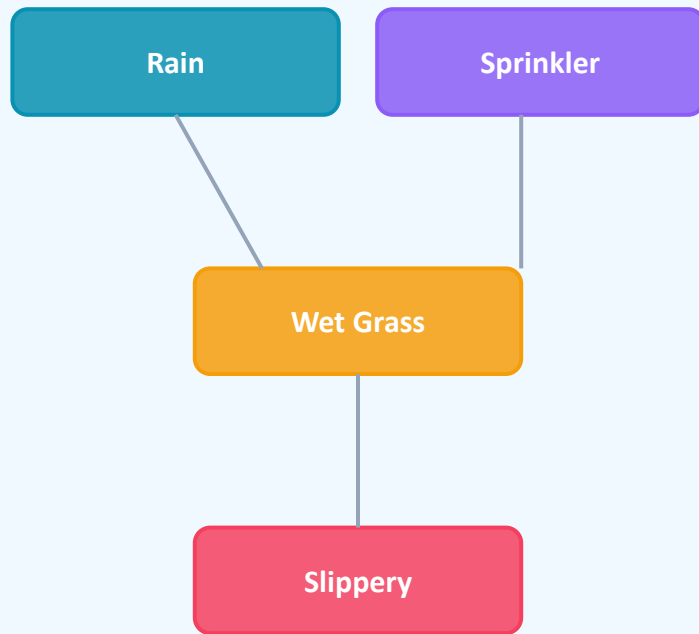
Random variables — observable or hidden

Directed Edges

Conditional dependencies between variables

CPTs

Conditional Probability Tables at each node



Directed Acyclic Graph (DAG)

Two Directions of Reasoning

Causal Reasoning ↓

Also called top-down or predictive reasoning. Flows with the arrows in the DAG.

"It rained heavily this morning."

- $P(\text{Wet Grass} \mid \text{Rain}) = 90\%$
- $P(\text{Slippery} \mid \text{Wet}) = 80\%$
- Downstream effects propagate

Diagnostic Reasoning ↑

Also called bottom-up or evidential reasoning. Flows against the arrows — uses Bayes' theorem.

"The grass is wet. Did it rain?"

- $P(\text{Rain} \mid \text{Wet Grass})$ updates
- Sprinkler probability also rises
- Evidence propagates upstream

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

Conditional Probability Tables (CPTs)

Each node stores a table of probabilities for every combination of parent states.

Rain	Sprinkler	$P(\text{Wet} = \text{Yes})$	$P(\text{Wet} = \text{No})$
Yes	Yes	0.99	0.01
Yes	No	0.90	0.10
No	Yes	0.80	0.20
No	No	0.01	0.99

Highlighted rows show high $P(\text{Wet} = \text{Yes})$ — strong causal influence from both parents.

Key Insight: The CPT allows the network to represent complex joint distributions compactly. Instead of storing all 2^n combinations, each node only stores its local conditional probabilities.

Real-World Applications



Medical Diagnosis

Symptoms $\rightarrow$ disease probability. Used in clinical decision support systems (e.g., QMR-DT for rare diseases).



Spam Filtering

Email words as feature nodes. $P(\text{Spam} \mid \text{'free', 'click'})$ — basis of early Bayesian spam classifiers.



Autonomous Vehicles

Sensor fusion: combine camera, LIDAR, radar readings to compute $P(\text{obstacle} \mid \text{all sensors})$.



Credit Risk Scoring

Income, history, debt ratio $\rightarrow P(\text{default})$. Banks use this for real-time loan decisions.



Legal & Forensics

DNA evidence, prior probability of guilt, test reliability $\rightarrow$ posterior probability of guilt.

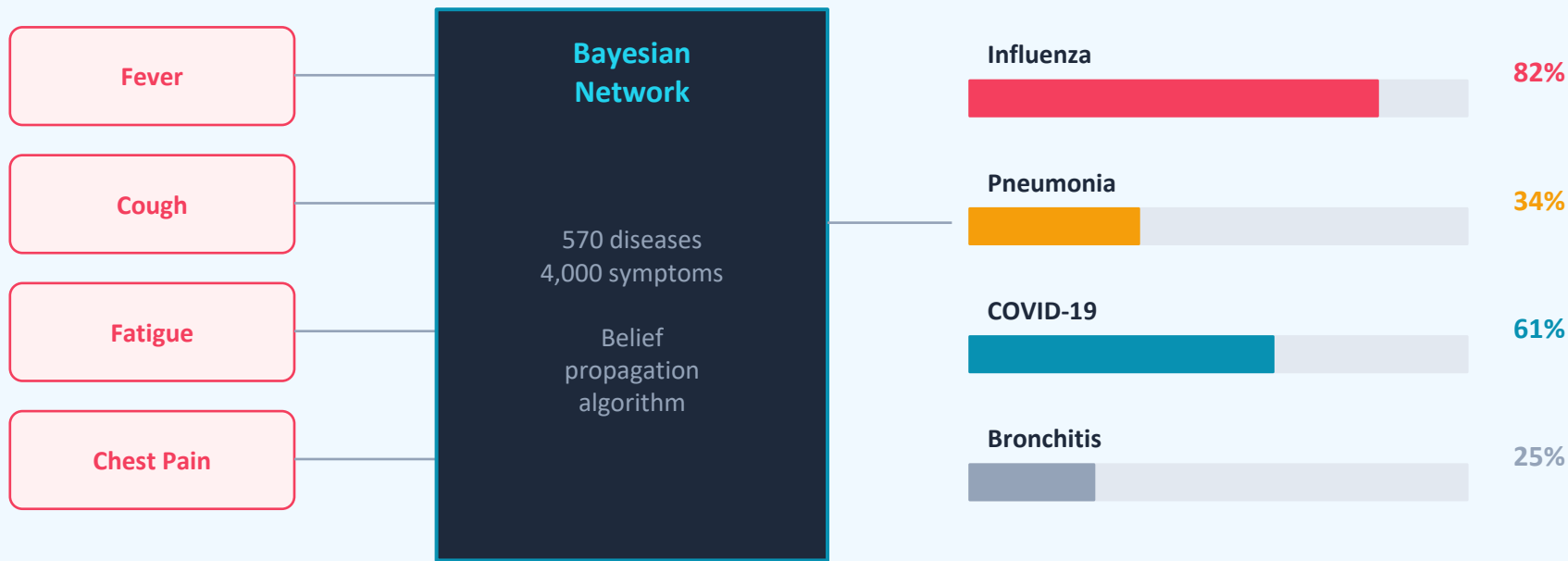


Fault Diagnosis

NASA uses Bayes nets to diagnose spacecraft faults from sensor readings in real-time.

Case Study: Medical Diagnosis

Bayesian networks were among the first AI systems deployed in clinical medicine. The QMR-DT (Quick Medical Reference - Decision Theoretic) system, developed at Carnegie Mellon, modeled 570 diseases and 4,000 symptoms.



Posterior probabilities after observing symptoms

Bayesian Networks in Large Language Models

Implicit Bayesian Structure in Attention

Transformer attention computes something analogous to posterior probabilities: $P(\text{next token} \mid \text{context})$. Softmax attention weights represent conditional beliefs over positions.

Retrieval-Augmented Generation (RAG)

RAG mirrors Bayesian updating: prior knowledge (pretrained weights) + new evidence (retrieved documents) $\rightarrow$ posterior answer. Evidence updates the model's beliefs at inference time.

Chain-of-Thought Reasoning

Step-by-step prompting replicates belief propagation: the model conditions each step on the previous, progressively narrowing uncertainty — just as evidence flows through a Bayes net.

Calibrated Uncertainty (Bayesian LLMs)

Explicit Bayesian neural networks output probability distributions over answers, not point estimates. This gives calibrated confidence: 'I'm 70% sure' — not just a single token prediction.

Attention Mechanism as Probabilistic Inference

The scaled dot-product attention formula can be interpreted as computing a posterior distribution over values, conditioned on the query matching keys.

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$$

Q (Query)

Bayesian role:

Prior belief / question about what to attend to

K (Keys)

Bayesian role:

Evidence — what's available in context

V (Values)

Bayesian role:

The posterior update — weighted information

softmax() normalizes scores into a proper probability distribution — making attention a learned conditional probability: $P(\text{attend to position } j \mid \text{query } i)$

Why We Need Approximate Inference

Exact inference in a Bayesian Network is computationally intractable for large, densely connected graphs. Here's why:

Exact inference complexity: $O(n \cdot 2^k)$ where n = nodes, k = max parent count

01 Exponential State Space

A network with 30 binary nodes has over 1 billion possible joint assignments. Summing over all hidden variables (marginalization) becomes computationally explosive.

02 NP-Hard in General Graphs

Exact inference is proven NP-hard for general Bayesian networks. Even approximate inference is NP-hard to guarantee within certain error bounds.

03 Real-World Networks Are Huge

Medical networks (thousands of diseases + symptoms), language models, and sensor fusion networks are far too large for exact variable elimination or belief propagation.

Solution: Approximate inference methods — Direct Sampling, MCMC, Variational Inference — trade exactness for scalability.

Direct Sampling

The simplest approximate inference method — generate random samples, count frequencies.

1

Order nodes topologically

Start from root nodes (no parents) and work down. The DAG guarantees a valid ordering always exists.

2

Sample each node from its CPT

For each node, look up its CPT conditioned on already-sampled parent values. Draw a random value from that distribution.

3

Repeat N times

Generate N complete joint samples across all variables. Each sample is one possible world consistent with the network's distributions.

4

Count to estimate probability

$P(\text{Query}) \approx (\text{samples where Query is true}) / N$. As $N \rightarrow \infty$, the estimate converges to the true probability.

Example: $P(\text{Rain}) \approx 647 / 1000 \text{ samples} \rightarrow \approx 65\%$

Limitation: Rejection sampling wastes samples on rare evidence events

Why Bayesian Networks Matter

01 Interpretability

Unlike black-box neural nets, the causal structure is explicit and readable by humans.

02 Handles Uncertainty

Natively represents degrees of belief — not just yes/no but a full probability distribution.

03 Bidirectional Inference

Reason forward (prediction) and backward (diagnosis) in the same model.

04 Small Data Efficiency

Encode domain knowledge in the graph structure — no need for millions of examples.

05 Causal Modeling

Supports counterfactual reasoning: 'What if X had been different?' — essential for decision-making.

06 Modular & Updateable

Add new nodes/edges as domain knowledge grows — no retraining everything from scratch.

Key Takeaways

- 1 Bayesian Networks model uncertainty through directed graphs with conditional probability tables
- 2 They enable both causal (predictive) and diagnostic reasoning in the same framework
- 3 Real-world impact spans medicine, finance, autonomous systems, law, and space exploration
- 4 LLMs use Bayesian-like reasoning implicitly — attention is learned conditional inference
- 5 RAG, chain-of-thought, and calibrated uncertainty are all Bayesian concepts in modern AI

"Probability is the language of uncertainty — and Bayesian networks are its grammar."